

Smart company IOT system

Our project Idea

At Smart IoT Company our vision is to revolutionize the modern workspace. We believe that any organization can be transformed into an intelligent, efficient, and secure environment by harnessing the power of the Internet of Things (IoT) and Artificial Intelligence (AI). Our mission is to integrate these technologies to build a smarter, more connected world, one business at a time.

Problem Statement

Many businesses today operate with outdated systems that create significant inefficiencies and security gaps. These challenges include:

- **Security Vulnerabilities:** Traditional security measures often lack the ability to accurately track who is in a building and at what times.
- **Wasted Resources:** Manual control of lighting, climate, and waste management leads to unnecessary energy consumption and operational costs.
- **Lack of Actionable Data:** Without a centralized system, data from different parts of a facility is often siloed, making it impossible to gain a holistic view of operations and make informed decisions.

The Solution: An Integrated IoT Ecosystem

Our solution is a comprehensive, integrated system that tackles these challenges head-on. We create a seamless network of smart devices that communicate and work together to automate and optimize the workplace environment.

The key components of our solution include:

- **Smart Attendance:** An advanced attendance system that uses RFID for initial authentication and an AI-powered camera for facial recognition, ensuring a secure and accurate log of everyone.
- **Automated Resource Management:**
 - Motion-Activated Lighting: An IR sensor detects movement and activates an LED, ensuring that lights are only on when needed.
 - Climate Control: A DHT11 sensor constantly monitors temperature and humidity, allowing for automated adjustments to maintain a comfortable and energy-efficient environment.
 - Smart Waste Management: An ultrasonic sensor measures the fullness of trash bins and sends alerts when they need to be emptied, optimizing collection schedules.
 - Auto lightening control: An LDR sensor to control the number of on lamps according to the surrounding light for better energy saving
 - Renewable energy resources: A solar cell panel used as an energy source
- **Smart Parking:** A system that uses IR sensors to detect available parking spaces, displaying the free space on an LCD screen and managing gate access with a servo motor.

All data from these services is sent to a central Communication Hub with MQTT protocol for real-time monitoring via Node-RED platform

Our prototype: -

